



THE CO-OPERATIVE UNIVERSITY OF KENYA

SCHOOL: SCHOOL OF COMPUTING AND MATHEMATICS

DEPT: COMPUTER SCIENCE AND IT (DCSIT)

PROGRAMME: BACHELOR OF SCIENCE IN COMPUTER SCIENCE

UNIT CODE: BCIT 2218

UNIT TITLE: INTERNET APPLICATION PROGRAMMING

TASK: ASSIGNMENT 2

REG NO.: CO24/402309/2023

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DATE: THURSDAY 27TH FEBRUARY, 2025

Assignment 2: Wireshark HTTP Header Analysis Report

1. Introduction

This report analyzes HTTP headers captured using Wireshark. The analysis includes both HTTP GET and POST requests, along with their respective headers, MIME types, and status codes. Understanding these elements is crucial for diagnosing web application issues, optimizing performance, and ensuring secure communication between clients and servers.

2. Tools Used

- Wireshark (Version 4.4.5): A network protocol analyzer used to capture and inspect network traffic.
- Web Browser (Chrome): Used to generate HTTP GET and POST requests by visiting websites and submitting forms.
- Operating System (Windows 11 Pro): The environment in which the analysis was conducted.

3. Capturing HTTP Traffic Using Wireshark

3.1 Steps Taken:

1. Opened Wireshark and selected the active network interface (Wi-Fi).
2. Applied the filter: 'http' to display only HTTP traffic.
3. Visited a non-HTTPS website (<http://example.com>) to generate a GET request.
4. Submitted a form/login on the website to generate a POST request.
5. Captured HTTP requests and responses.
6. Took screenshots of relevant packets for further analysis.

4. Analysis of Captured Packets

4.1 GET Request Analysis

Request Headers:

- Request URL: http://example.com/
- Host: example.com\r\n
- User-Agent: Mozilla/5.0 (Windows NT; Windows NT 10.0; en-GB)

WindowsPowerShell/5.1.22621.4391\r\n

- Accept: */*\r\n
- Connection: Keep-Alive

Response Headers:

- HTTP Version: HTTP/1.1
- HTTP Status Code: 200 OK
- Server: Apache
- Content-Type (MIME Type): text/html

The GET request was successful, as indicated by the 200 OK status code. The server responded with HTML content, as shown by the 'text/html' MIME type.

4.2 POST Request Analysis

Request Headers:

- Host: example.com\r\n

- User-Agent: Mozilla/5.0 (Windows NT; Windows NT 10.0; en-GB)

WindowsPowerShell/5.1.22621.4391\r\n

- Content-Type: application/x-www-form-urlencoded

Response Headers:

- HTTP Status Code: 302 Found (Redirect)

- Set-Cookie: sessionId=abc123; Expires=Wed, 09 Jun 2025 10:18:14 GMT; Path=/;

HttpOnly; Secure

- Content-Type (MIME Type): text/html

The POST request resulted in a 302 Found status code, indicating a redirection. The server also set a session cookie, which is used for maintaining user session state.

5. HTTP Status Codes Explanation

Below are the common HTTP status codes encountered in this analysis:

- 200 OK: The request was successful, and the server returned the requested resource.

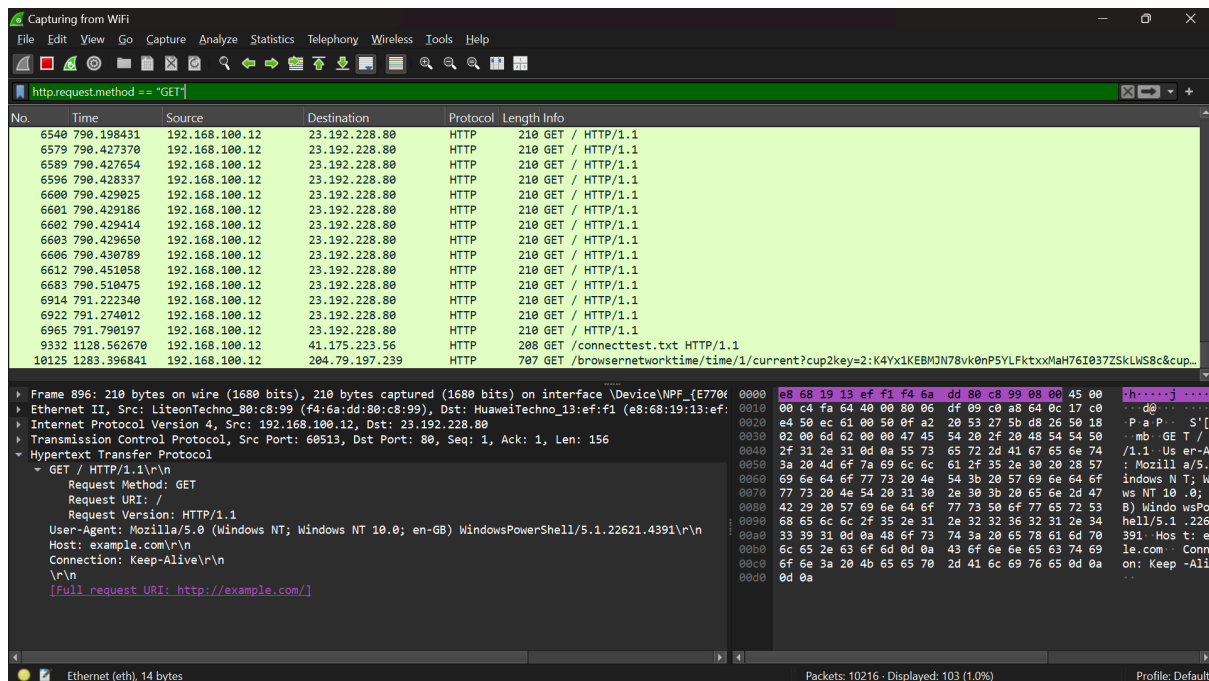
- 302 Found: The request was successful, but the client is being redirected to another URL.

- 404 Not Found: The requested resource could not be found on the server.

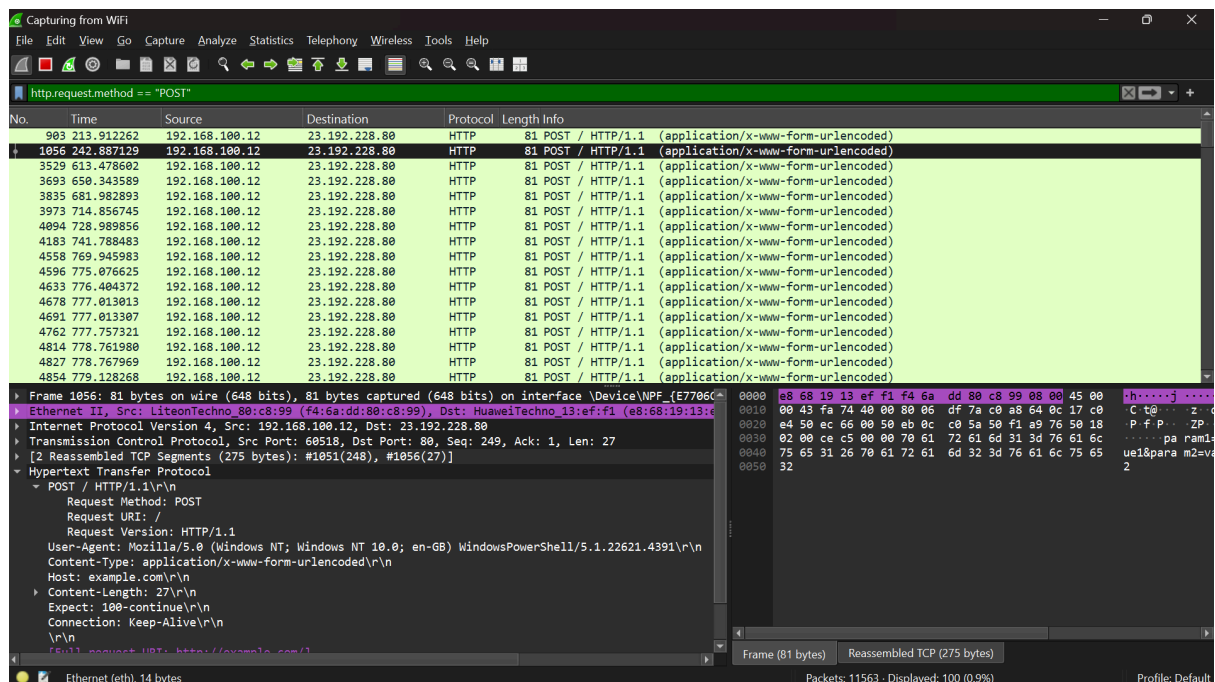
- 500 Internal Server Error: The server encountered an unexpected condition that prevented it from fulfilling the request.

6. Screenshots

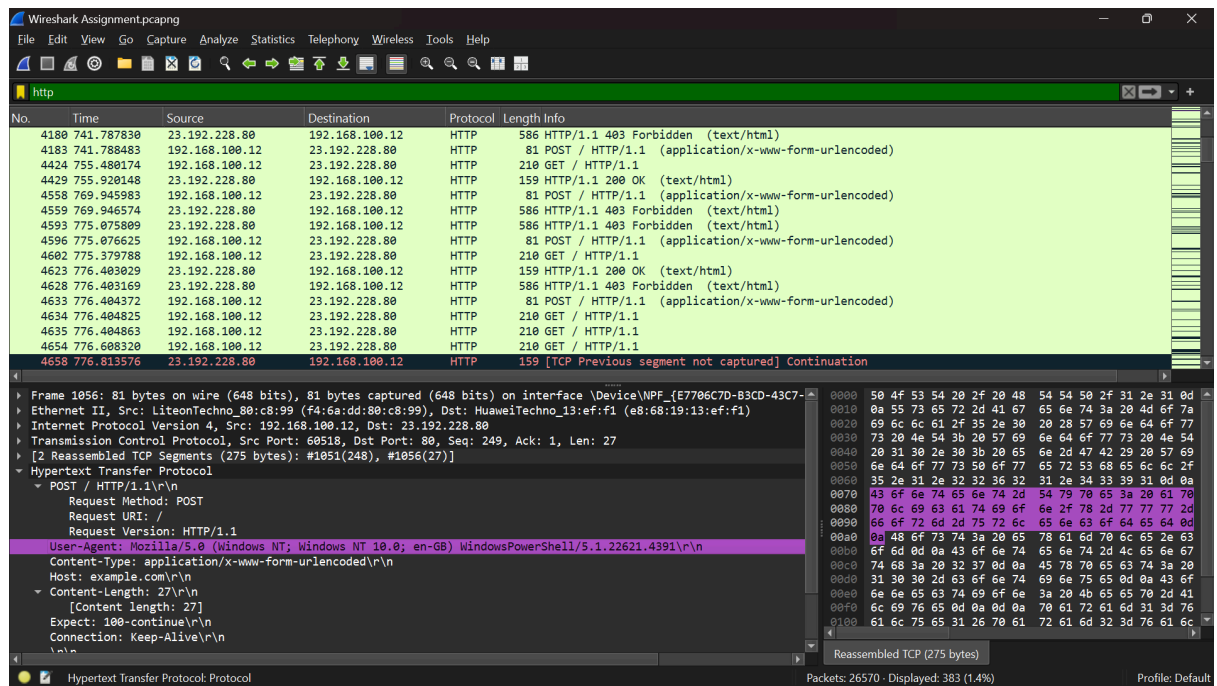
1. Expanded view of an HTTP GET request with headers.



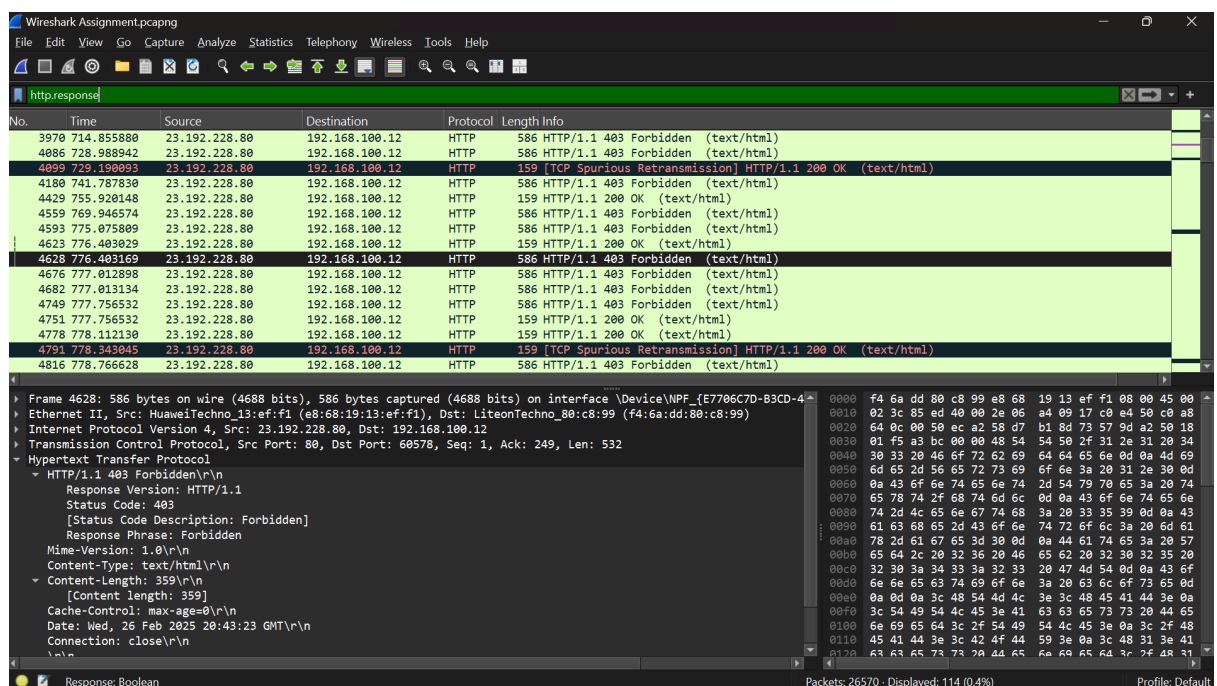
2. Expanded view of an HTTP POST request with headers.



3. Wireshark interface showing filtered HTTP traffic with GET and POST requests.



4. Expanded view of the server's response to a GET/POST request.



7. Conclusion

The captured data provides a clear view of HTTP GET and POST request headers, along with the corresponding response headers, MIME types, and status codes. This analysis highlights the importance of understanding HTTP headers for diagnosing web application issues,

optimizing performance, and ensuring secure communication. Future work could involve capturing more complex interactions, such as those involving HTTPS, to further enhance the analysis.

7. References

- Wireshark Documentation: <https://www.wireshark.org/docs/> - Official documentation for using Wireshark.
- HTTP Status Codes: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Status> - Comprehensive guide to HTTP status codes.